RESEARCH DIRECTION

Power-Aware HBM/HBF Co-Design for Disaggregated MoE Serving

Yu-Jui Liang Advisor: Hsiang-Yun Cheng
National Taiwan University / Academia Sinica

Research Meeting 11 November 2026

Proposed research direction

Proposed Research Direction

Hardware Heterogeneous GPU HBM/HBF configurations.

Serving Attention/Expert separation, memory residency and scheduling.

Objective Improve energy efficiency under latency/SLO and hardware constraints.

Evaluation metrics

Metric	Unit
Energy per output token	J/token
Energy efficiency	tokens/J
Time to first token (TTFT)	ms
Time per output token (TPOT)	ms/token
Goodput under SLO	tokens/s
Operational carbon emissions per output token	gCO_2e/token

Research goal: Reduce power consumption and carbon emissions across data centers through hardware co-design and coordinated serving, while meeting latency and SLO requirements.

Why Heterogeneous HBM/HBF GPUs?

Data	Access behavior	Candidate placement
Active KV cache	Frequent decode reads, incremental updates. Latency/bandwidth sensitive.	HBM is a natural baseline.
Hot expert weights	Read-mostly with frequent reuse.	HBM cache may help.
Cold expert weights	Large capacity, read-mostly.	HBF may add residency capacity.
Intermediate activations	Transfer between GPU workers.	Explicit network cost and synchronization.

Research question

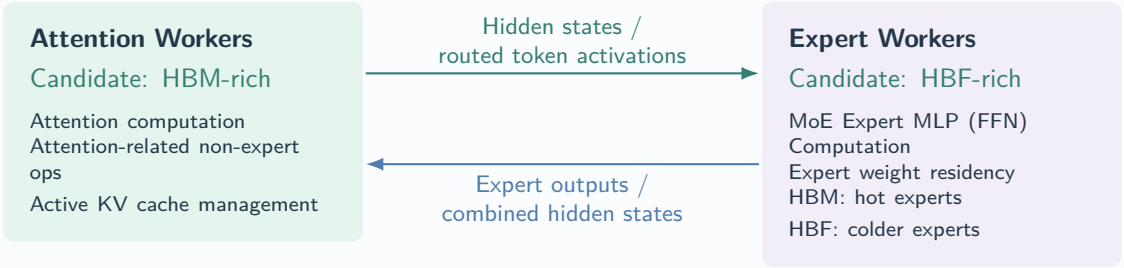
Can role-specific HBM/HBF configurations improve the energy efficiency of MoE serving?

Capacity is a possible benefit. HBF reads and GPU waiting can consume it.

HBF does not automatically save power.

Proposed Attention/Expert Architecture

Hypothesis to evaluate Within one data center.

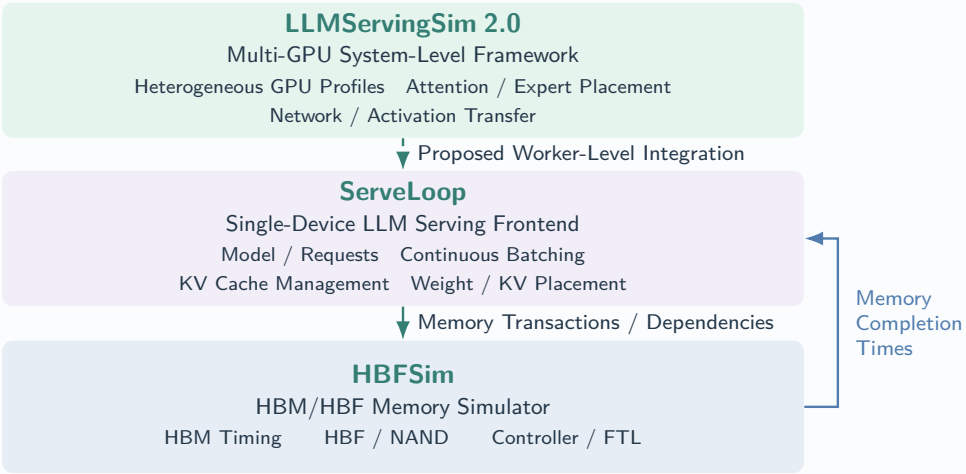


Residency: We do NOT transfer the entire historical KV cache or all expert weights between worker groups at every layer.

Costs remain

Activation communication, synchronization and HBF read stalls may dominate performance.

Proposed Simulation Integration Stack



Goal: Evaluate heterogeneous HBM/HBF GPU configurations for Attention–Expert MoE serving.

Proposed architecture; multi-GPU integration and energy-model calibration remain future work.

Research Questions and Next Steps

RQ1 Does HBM-rich Attention + HBF-rich Experts beat homogeneous or monolithic alternatives in energy efficiency under the same SLO constraints?

RQ2 When do HBF read latency, GPU waiting energy and A/E communication outweigh residency benefits?

RQ3 Which configurations give the best energy–goodput trade-off under comparable capacity, package, cost and power constraints?

NEXT WEEK — November 18, 2026

MemExplorer: Navigating the Heterogeneous Memory Design Space for Agentic Inference NPUs

Already explores heterogeneous compute/memory, including HBM/HBF, and energy/performance trade-offs. Study its actual novelty coverage and limitations before finalizing our contribution.

A separate paper-review presentation next week.

Related Work Comparison Matrix

Paper	Simulator / Evaluation	Cross Data Center	Partition / Placement	Objective / (Eval. Metrics)
Helix	Own sim. + GPU cluster	Yes (sim.)	Layers, pipeline, routing	Throughput, latency
ThunderServe	Estimator + GPU tests	Not shown	P/D, TP/PP; KV	Throughput, perf/$
Hetis	vLLM prototype; sim. NV	Not shown	Attention heads; KV	Throughput, KV use
Tessera	GPU runtime; sim. NV	Not shown	Kernel placement	Throughput, perf/$
GreenCache	Prototype + replay/model	No	SSD KV/prefix placement	Carbon + SLO
Janus	Serving runtime; sim. NV	Not shown	A/E pools; experts	Per-GPU throughput
How Far Can Disaggregation Go?	Profiles + ASTRA-Sim	Not shown	A/FFN; GPU placement	SLO throughput
Carbon–SLO	Geo queue/network sim.	Yes	Request / batch routing	Carbon + SLO
LLMServingSim 2.0	Multi-GPU sim. + profiles	Not shown	Heterogeneous TP/PP/EP	(Perf., energy, fidelity)
HBFSim / ServeLoop	Event-driven / one device	No	Transactions; KV/weights	(Timing, wear/thermal)
MemExplorer	Analytical co-design DSE	Not shown	HBM/HBF; P/D, A/FFN	Token/J, performance
Characterizing HBF	Trace-driven request sim.	Not shown	Multi-GPU; KV/experts	(Time, energy, wear)
Potential Applications of HBF	Capacity/replication model	Not shown	Expert residency/replicas	(TPOT, network traffic)